PPC TENSOR EQUATION AND MERCURY'S PERIHELION PRECESSION

Claim A: Recovery of the General-Relativistic Vacuum Prediction

Research Paper

Prepared from the PPC Tensor Equations and PPC Weak-Field Equations formulation

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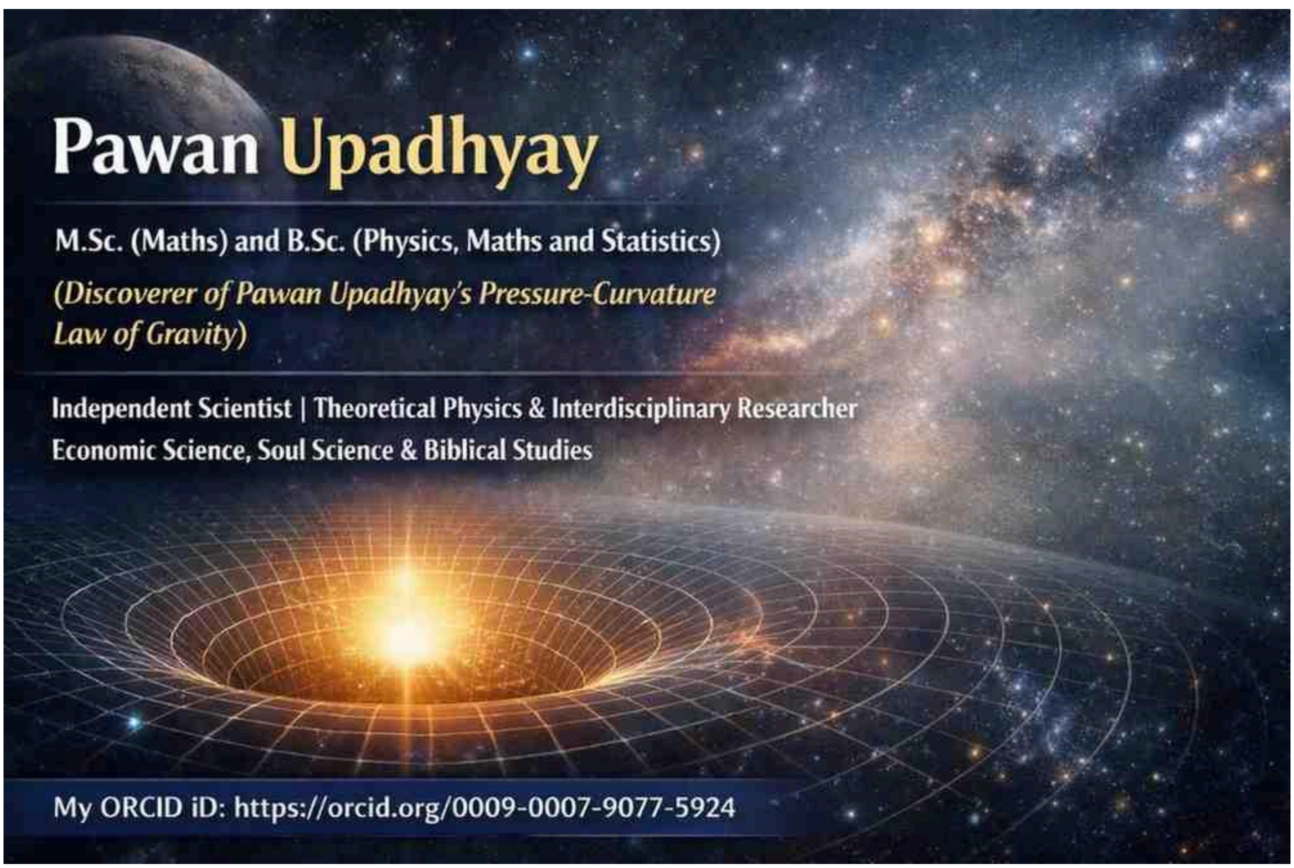

Abstract.

This paper examines Claim A of the proposed PPC Law of Gravity: that the present PPC tensor equation, when applied to the vacuum exterior of a spherically symmetric gravitating body, recovers the vacuum Einstein equation and consequently the Schwarzschild exterior. The PPC formulation introduces gravitational pressure P_g through the equation of state P_g = omega E_d, with E_d = rho c^2, and uses a perfect-fluid stress-energy form. In vacuum, E_d = P_g = 0, so the PPC tensor equation reduces to G_mu_nu = 0. The static spherical exterior is therefore Schwarzschild, and the corresponding test-particle orbit contains the relativistic 3GM u^2/c^2 correction. This yields the standard perihelion advance Delta phi = 6 pi GM/[a(1-e^2)c^2], approximately 43 arcseconds per century for Mercury. The result establishes that the present PPC tensor formulation reproduces the GR vacuum prediction. It does not by itself establish an independent PPC mechanism for the anomalous precession; that would require a separate derivation showing how the PPC pressure-field formulation generates the post-Newtonian orbital correction.

1. Introduction

The PPC framework developed in the accompanying formulation treats gravitational pressure P_g as a fundamental gravitational-pressure variable. The supplied PPC document describes P_g as the effective pressure contributing to spacetime curvature and gives a perfect-fluid tensor formulation in which the pressure variable is P_g. It also presents the weak-field relation in which E_d + 3P_g controls curvature strength and the pressure gradient is associated with the geometry or path followed by matter.

The purpose of this paper is deliberately limited. It addresses Claim A only: whether the present PPC tensor equation has the same vacuum limit as the Einstein field equation and, therefore, whether it reproduces the Schwarzschild exterior and the standard relativistic Mercury perihelion shift.

2. PPC Matter, Energy Density, and Gravitational Pressure

E_d(x,y,z,ct) = rho(x,y,z,ct) c^2

P_g(x,y,z,ct) = omega(x,y,z,ct) E_d(x,y,z,ct)

Here rho is taken, according to the PPC formulation used in this work, as the total matter mass density field, E_d is its corresponding energy density, and omega is the equation-of-state parameter. The gravitational pressure is therefore not introduced as an unrelated variable but through the PPC equation of state.

3. PPC Perfect-Fluid Tensor Equation

T_munu = (E_d + P_g) u_mu u_nu + P_g g_munu

G_munu = (8 pi G / c^4) T_munu

The PPC document presents this perfect-fluid tensor structure as its gravitational-field equation, with P_g replacing the ordinary perfect-fluid pressure variable. Consequently, after using P_g = omega E_d and E_d = rho c^2, the equation can be written as

G_munu = (8 pi G / c^2) rho [(1 + omega) u_mu u_nu + omega g_munu]

4. Vacuum Limit of the PPC Tensor Equation

The Mercury perihelion calculation concerns the solar exterior. For the local vacuum region outside the Sun, the matter-source fields vanish:

$$\text{rho} = 0, \quad \text{E\_d} = 0, \quad \text{P\_g} = 0$$

The PPC stress-energy tensor consequently becomes

$$\text{T\_munu} = 0$$

and the PPC tensor field equation reduces to

$$\text{G\_munu} = 0$$

This is the vacuum Einstein equation. Therefore the static, spherically symmetric asymptotically flat exterior solution is the Schwarzschild metric.

5. Schwarzschild Exterior

$$\text{ds\^{}2} = -(1 - \text{2GM}/(\text{c\^{}2 r}))\ \text{c\^{}2 dt\^{}2} + (1 - \text{2GM}/(\text{c\^{}2 r}))\text{\^{}(-1) dr\^{}2} + \text{r\^{}2 dOmega\^{}2}$$

The mass parameter M is determined by the gravitating source and the chosen asymptotic boundary conditions. The important point for Claim A is that the PPC matter and gravitational-pressure terms disappear from the local vacuum field equation, leaving the same vacuum geometry.

6. Relativistic Orbital Equation

For a test body following a timelike geodesic in the Schwarzschild exterior, restrict the motion to the equatorial plane and define $u = 1/r$. The orbit equation can be written, to first post-Newtonian order, as

$$\text{d\^{}2u/dphi\^{}2} + \text{u} = \text{GM/h\^{}2} + (\text{3GM/c\^{}2})\ \text{u\^{}2}$$

The first term on the right is the Newtonian contribution. The second is the leading relativistic correction. It changes the radial oscillation frequency relative to the azimuthal frequency and therefore prevents the orbit from closing after one radial cycle.

7. Perihelion Advance

For a nearly Keplerian orbit with semi-major axis a and eccentricity e, perturbation theory gives the advance of perihelion per revolution:

$$\text{Delta phi} = \text{6 pi GM} / [\text{a } (1 - \text{e\^{}2})\ \text{c\^{}2}]$$

For Mercury, using its measured orbital scale and eccentricity gives approximately 0.1035 arcseconds per revolution. With roughly 415.2 Mercury revolutions per century, this corresponds to approximately

$$\text{Delta phi\_Mercury} \approx 43 \text{ arcseconds per century}$$

This is the well-known relativistic contribution to Mercury's perihelion advance after the classical contributions are accounted for.

8. Claim A: Result

PPC vacuum tensor equation -> G_munu = 0 -> Schwarzschild exterior -> relativistic perihelion advance

Claim A is therefore supported by the present PPC tensor formulation, in the following precise sense: because the PPC tensor equation reduces to the vacuum Einstein equation when E_d and P_g vanish, its static spherical exterior is the Schwarzschild exterior and its test-particle prediction includes the standard 43 arcseconds-per-century Mercury perihelion effect.

9. Scope and Scientific Qualification

The conclusion above should not be interpreted as a demonstration that PPC has independently derived the Mercury anomaly from a new gravitational mechanism. The tensor equation used here has the Einstein perfect-fluid tensor structure with the PPC gravitational pressure P_g as the pressure variable. Thus the vacuum prediction follows from the shared vacuum equation G_munu = 0.

A stronger and distinct PPC claim would require a separate derivation beginning with the PPC fields P_g(x,y,z,ct), E_d(x,y,z,ct), and omega(x,y,z,ct), together with the PPC force-density relation F = rho g = -nabla P_g, and then deriving the post-Newtonian metric and orbital equation. In particular, one would need to derive the coefficient 3GM/c^2 in the u^2 term rather than insert it from the known Schwarzschild result.

10. Conclusion

The present PPC tensor formulation reproduces the GR vacuum result. In the solar exterior, where the local matter energy density and gravitational pressure vanish, the PPC tensor equation becomes G_munu = 0. Its static, spherical, asymptotically flat solution is the Schwarzschild geometry. The resulting test-particle orbit has the relativistic correction 3GM u^2/c^2 and consequently predicts approximately 43 arcseconds per century for Mercury's perihelion advance. Therefore Claim A is established as a recovery of the GR vacuum prediction. The independent PPC mechanism, if intended, remains a separate question for future derivation.

References

[1] PPC Tensor Equations and PPC weak field equations.pdf. Supplied PPC formulation used as the primary source for the equations and terminology in this paper.

[2] A. Einstein, “Die Feldgleichungen der Gravitation,” Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften (Berlin), 1915.

[3] A. Einstein, “Erklärung der Perihelbewegung des Merkur aus der allgemeinen Relativitätstheorie,” Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften (Berlin), 1915.

Source note: The PPC-specific equations and terminology above are based on the user's supplied PPC document. The Schwarzschild exterior and standard relativistic perihelion result are used here as the mathematical consequence of the vacuum Einstein equation.

In Print Style :

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RESEARCH PAPER

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ABSTRACT

This paper examines Claim A of the proposed PPC Law of Gravity: that the present PPC tensor equation, when applied to the vacuum exterior of a spherically symmetric gravitating body, recovers the vacuum Einstein equation and consequently the Schwarzschild exterior.

The PPC formulation introduces gravitational pressure Pg through the equation of state:

$P_g = \omega E_d$

with:

$Ed\_ = \rho c^2$

and uses a perfect-fluid stress-energy form.

In vacuum:

$E_d = P_g = 0$

so the PPC tensor equation reduces to:

$G_{\mu\nu} = 0$

The static spherical exterior is therefore Schwarzschild, and the corresponding test-particle orbit contains the relativistic:

$3GMu^2/c^2$

correction.

This yields the standard perihelion advance:

$\Delta\varphi = 6\pi GM / [a(1 - e^2)c^2]$

approximately 43 arcseconds per century for Mercury.

The result establishes that the present PPC tensor formulation reproduces the GR vacuum prediction. It does not by itself establish an independent PPC mechanism for the anomalous precession; that would require a separate derivation showing how the PPC pressure-field formulation generates the post-Newtonian orbital correction.

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1. INTRODUCTION

The PPC framework developed in the accompanying formulation treats gravitational pressure P_g as a fundamental gravitational-pressure variable.

The supplied PPC document describes P_g as the effective pressure contributing to spacetime curvature and gives a perfect-fluid tensor formulation in which the pressure variable is P_g. It also presents the weak-field relation in which $E_d + 3P_g$ controls curvature strength and the pressure gradient is associated with the geometry or path followed by matter.

The purpose of this paper is deliberately limited. It addresses Claim A only: whether the present PPC tensor equation has the same vacuum limit as the Einstein field equation and, therefore, whether it reproduces the Schwarzschild exterior and the standard relativistic Mercury perihelion shift.

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2. PPC MATTER, ENERGY DENSITY, AND GRAVITATIONAL PRESSURE

$\mathbf{E_d(x,y,z,ct) = \rho(x,y,z,ct)c^2}$

$\mathbf{P_g(x,y,z,ct) = \omega(x,y,z,ct)E_d(x,y,z,ct)}$

Here ρ is taken, according to the PPC formulation used in this work, as the total matter mass density field, E_d is its corresponding energy density, and ω is the equation-of-state parameter.

The gravitational pressure is therefore not introduced as an unrelated variable but through the PPC equation of state.

3. PPC PERFECT-FLUID TENSOR EQUATION

$$T_{\mu\nu} = (E_d + P_g)u_\mu u_\nu + P_g g_{\mu\nu}$$

$$G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$$

The PPC document presents this perfect-fluid tensor structure as its gravitational-field equation, with Pg replacing the ordinary perfect-fluid pressure variable.

Using:

$P_g = \omega E_d$

and:

$E_d = \rho c^2$

the equation can be written as:

$$G_{\mu\nu} = (8\pi G/c^2)\rho[(1 + \omega)u_\mu u_\nu + \omega g_{\mu\nu}]$$

4. VACUUM LIMIT OF THE PPC TENSOR EQUATION

The Mercury perihelion calculation concerns the solar exterior.

For the local vacuum region outside the Sun, the matter-source fields vanish:

$\rho = 0$

$E_d = 0$

$P_g = 0$

The PPC stress-energy tensor consequently becomes:

$T_{\mu\nu} = 0$

and the PPC tensor field equation reduces to:

$G_{\mu\nu} = 0$

This is the vacuum Einstein equation.

Therefore, the static, spherically symmetric, asymptotically flat exterior solution is the Schwarzschild metric.

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5. SCHWARZSCHILD EXTERIOR

$$ds^2 = -(1 - 2GM/c^2r)c^2dt^2$$
$$+ (1 - 2GM/c^2r)^{-1}dr^2$$
$$+ r^2d\Omega^2$$

The mass parameter M is determined by the gravitating source and the chosen asymptotic boundary conditions.

The important point for Claim A is that the PPC matter and gravitational-pressure terms disappear from the local vacuum field equation, leaving the same vacuum geometry.

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6. RELATIVISTIC ORBITAL EQUATION

For a test body following a timelike geodesic in the Schwarzschild exterior, restrict the motion to the equatorial plane and define:

$u = 1/r$

The orbit equation can be written, to first post-Newtonian order, as:

$d^2u/d\varphi^2 + u = GM/h^2 + (3GM/c^2)u^2$

The first term on the right is the Newtonian contribution.

The second term is the leading relativistic correction.

It changes the radial oscillation frequency relative to the azimuthal frequency and therefore prevents the orbit from closing after one radial cycle.

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7. PERIHELION ADVANCE

For a nearly Keplerian orbit with semi-major axis a and eccentricity e, perturbation theory gives the advance of perihelion per revolution:

$$\Delta\varphi = 6\pi GM / [a(1 - e^2)c^2]$$

For Mercury, using its measured orbital scale and eccentricity gives approximately:

0.1035 arcseconds per revolution

With roughly 415.2 Mercury revolutions per century, this corresponds to approximately:

$\Delta\varphi_{(\text{Mercury})} \approx 43$ arcseconds per century

This is the well-known relativistic contribution to Mercury's perihelion advance after the classical contributions are accounted for.

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8. CLAIM A: RESULT

The logical sequence is:

PPC vacuum tensor equation
↓
$G_{\mu\nu} = 0$
↓
Schwarzschild exterior
↓
Relativistic perihelion advance

Claim A is therefore supported by the present PPC tensor formulation, in the following precise sense:

Because the PPC tensor equation reduces to the vacuum Einstein equation when E_d and P_g vanish, its static spherical exterior is the Schwarzschild exterior and its test-particle prediction includes the standard 43 arcseconds-per-century Mercury perihelion effect.

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9. SCOPE AND SCIENTIFIC QUALIFICATION

The conclusion above should not be interpreted as a demonstration that PPC has independently derived the Mercury anomaly from a new gravitational mechanism.

The tensor equation used here has the Einstein perfect-fluid tensor structure with the PPC gravitational pressure P_g as the pressure variable. Thus the vacuum prediction follows from the shared vacuum equation:

$$G_{\mu\nu} = 0$$

A stronger and distinct PPC claim would require a separate derivation beginning with the PPC fields:

$P_g(x,y,z,ct)$

$E_d(x,y,z,ct)$

$\omega(x,y,z,ct)$

together with the PPC force-density relation:

$$F = \rho g = -\nabla P_g$$

and then deriving the post-Newtonian metric and orbital equation.

In particular, one would need to derive the coefficient:

$$3GM/c^2$$

in the u^2 term rather than insert it from the known Schwarzschild result.

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10. CONCLUSION

The present PPC tensor formulation reproduces the GR vacuum result.

In the solar exterior, where the local matter energy density and gravitational pressure vanish, the PPC tensor equation becomes:

$$G_{\mu\nu} = 0$$

Its static, spherical, asymptotically flat solution is the Schwarzschild geometry.

The resulting test-particle orbit has the relativistic correction:

$$3GMu^2/c^2$$

and consequently predicts approximately:

43 arcseconds per century

for Mercury's perihelion advance.

Therefore Claim A is established as a recovery of the GR vacuum prediction.

The independent PPC mechanism, if intended, remains a separate question for future derivation.

REFERENCES

[1] *PPC Tensor Equations and PPC Weak Field Equations.pdf.* Supplied PPC formulation used as the primary source for the equations and terminology in this paper.

[2] A. Einstein, "Die Feldgleichungen der Gravitation," *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften (Berlin)*, 1915.

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Source Note: The Schwarzschild exterior and standard relativistic perihelion result are used here as the mathematical consequence of the vacuum Einstein equation.

SPECIAL NOTE

In the PPC formulation, the symbol ρ represents the **mass density field**, written as:

$\rho(x, y, z, ct)$

The term **mass density** refers to the amount of mass per unit volume at a particular location, whereas **mass density field** refers to the complete spatial and temporal distribution of mass density throughout the physical system.

Thus, in the PPC equations, ρ should preferably be described as a **mass density field**, because its value may vary with position and time.

Accordingly:

ρ = mass density at a specified point

$\rho(x, y, z, ct)$ = mass density field throughout spacetime

This terminology is consistent with the PPC formulation used in the paper, where:

$E_d(x, y, z, ct) = \rho(x, y, z, ct)c^2$

and ρ is identified as the total matter mass density field.

Important clarification

The use of the term **“mass density field”** does not mean that mass itself is necessarily a continuous substance. It specifies the mathematical description of how mass density is distributed over space and time.

For the PPC framework, the more precise terminology is therefore:

“total matter mass density field $\rho(x, y, z, ct)$”

rather than simply:

“mass density ρ.”

1. Mass density

Mass density is the amount of mass contained per unit volume.

It is usually written as:

$\boldsymbol{\rho = m / V}$

Unit: **kg/m^3**

For example, if a material has a density of 1,000 kg/m^3, that tells us how much mass exists per cubic metre.

2. Mass density field

A **mass density field** means that the density is treated as a function of position—and possibly time.

Instead of simply writing:

$\boldsymbol{\rho}$

I write:

$\boldsymbol{\rho(x, y, z, t)}$

This means the density can be different at different locations and at different times.

For example:

- At point A: $\rho = 5\ kg/m^3$
- At point B: $\rho = 2\ kg/m^3$
- At point C: $\rho = 0\ kg/m^3$

Together, these values form a **mass density field**.

In PPC paper

My paper specifically uses:

$$\mathbf{Ed(x,y,z,ct) = \rho(x,y,z,ct)c^2}$$

and describes ρ as the **total matter mass density field**.

That wording is appropriate because PPC is describing matter throughout spacetime, rather than assigning one single density value to the entire system.

A useful distinction is:

Mass density = a physical quantity at a particular location.

Mass density field = the complete distribution of that quantity throughout space (and possibly time).

So for my PPC formulation, I consistently using **"mass density field"** when referring to ρ(x,y,z,ct), and **"mass density"** when referring to its value at a particular point.

SPECIAL NOTE

The Meaning of Vacuum in the PPC Formulation

In the present PPC analysis, the term **"vacuum"** should be understood as an **idealized source-free region** outside the gravitating body.

For the purpose of Claim A, the local matter mass density field is taken to vanish:

$$\boldsymbol{\rho(x, y, z, ct) = 0}$$

Consequently, according to the PPC definitions,

$$\mathbf{E_d = \rho c^2 = 0}$$

and

$$\mathbf{P_g = \omega E_d = 0}$$

Therefore,

$$\mathbf{T_{\mu\nu} = 0}$$

and the PPC tensor equation reduces to:

$$\mathbf{G_{\mu\nu} = 0}$$

This is the vacuum equation used to obtain the Schwarzschild exterior and the standard relativistic perihelion correction.

Physical Qualification

The statement **"vacuum" does not necessarily mean that physical space is absolutely devoid of every possible form of energy or field**.

A physical region of space may, in principle, contain residual matter, radiation, electromagnetic fields, background energy, or other contributions to stress-energy. If such contributions are relevant to the region being studied, they should be included in the appropriate field equation rather than simply setting the total source to zero.

Therefore, the PPC Claim A calculation should be understood specifically as the **classical source-free vacuum approximation**:

$\rho = 0$, $E_d = 0$, $P_g = 0$

rather than as a claim that absolutely no physical energy can exist in empty space.

Important Consequence for PPC

If the PPC theory proposes that an additional gravitational-pressure or energy field remains present even where ordinary matter density is zero, then the assumption

$\rho = 0 \rightarrow E_d = 0 \rightarrow P_g = 0$

would no longer automatically follow.

In that case, the PPC theory would need to specify the independent vacuum behavior of its fields and derive the corresponding exterior metric and orbital equation.

This distinction is important because the present Claim A establishes recovery of the **standard GR vacuum result**, while an independent PPC prediction would require showing whether PPC possesses additional nonzero fields in the exterior region and, if so, how those fields modify the gravitational geometry.

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In an ideal GR vacuum

When I write

$\rho = 0$, $E_d = 0$, $P_g = 0$

I mean an **idealized classical vacuum region**: no matter or other gravitating energy-momentum is being included in that region. In GR, the vacuum Einstein equation is then

$G_{\mu\nu} = 0$.

But **“vacuum” does not necessarily mean that physical reality contains absolutely nothing whatsoever**. Einstein Online describes vacuum GR as a region with no matter and no other forms of energy, while real laboratory vacua can still contain extremely small amounts of residual matter.

What about energy in empty space?

There are several possibilities:

1. **Residual matter/radiation**
   A real region of space can contain extremely small amounts of gas, photons, particles, etc. Then its energy density is not exactly zero.

2. **Gravitational field itself**
   In GR, gravitational effects do not require local matter to be present. For example, the exterior of a star can be a vacuum region while spacetime is still curved. The Schwarzschild solution is precisely such a vacuum exterior.

3. **Vacuum energy / cosmological constant**
   Modern cosmology allows a vacuum-associated energy component, commonly represented in GR through the cosmological constant. This is important cosmologically, although it is not normally included in the simple Schwarzschild vacuum calculation.

This matters for PPC equation

Solar exterior → $\rho = 0$ → $E_d = 0$ → $P_g = 0$ → $G_{\mu\nu} = 0$

That is mathematically valid **provided “vacuum” is defined as the classical source-free vacuum used in the Schwarzschild derivation**.

Special Note on Vacuum: In this paper, “vacuum” refers to the idealized classical vacuum exterior used in deriving the Schwarzschild solution, in which the local PPC matter-energy density and gravitational pressure are taken to vanish: $\rho = 0$, $E_d = 0$, and $P_g = 0$. This does not imply that physical space is necessarily absolutely devoid of every possible form of energy or field. Any nonzero background energy density, radiation, electromagnetic field, cosmological constant, or other stress-energy contribution would require its inclusion in the field equation and could modify the exact vacuum condition.

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